A proof of the Elizalde–Luo conjecture for nonnesting permutations avoiding 1132 and 3312

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Abstract

Nonnesting permutations are permutations of the multiset $\{1, 1, 2, 2, \ldots, n, n\}$ that avoid the patterns 1221 and 2112; there are $n!\,C_n$ of them, where C_n is the nth Catalan number. In the concluding section of their systematic study of pattern avoidance in nonnesting permutations [DMTCS 27:1 (2025), #13], Elizalde and Luo conjectured, on the basis of data for $n \leq 8$, that the number of nonnesting permutations of $\{1, 1, \ldots, n, n\}$ avoiding both 1132 and 3312 equals $3^n - 3 \cdot 2^{n-1} + 1$ (OEIS A168583, shifted). We prove this conjecture. The proof is elementary and self-contained: nonnesting words factor as a Dyck shape carrying a label permutation; avoidance of $\{1132, 3312\}$ forces the labels to form a prefix-interval permutation, encoded by a sign word in $\{\mathsf{L}, \mathsf{H}\}^{n-1}$, and reduces to a two-coloring condition on the crossing arcs of the shape. The shapes admitting a valid coloring fall into three explicit families, and summing the number of colorings gives the formula. We also give a second, bijective proof of the final count: an explicit letter-by-letter bijection between the avoiders and a ternary language $W_n \subseteq \{\mathsf{A}, \mathsf{B}, \mathsf{C}\}^n$ whose cardinality is $3^n - 3 \cdot 2^{n-1} + 1$ by inspection. Every lemma has been verified mechanically far beyond the conjecture's original range, including an exhaustive word-level check of the structural characterization over all 57,657,600 nonnesting words for $n = 8$.

1 Introduction

A *Stirling permutation* [6] is a permutation of the multiset $[n]_2 = \{1, 1, 2, 2, \ldots, n, n\}$ in which the letters between the two copies of any value are larger than that value. Two natural relaxations have attracted attention in recent years: *noncrossing* (or *quasi-Stirling*) permutations, the permutations of $[n]_2$ with no subsequence *abab* with $a \neq b$, introduced by Archer, Gregory, Pennington and Slayden [1] and studied further in [3]; and *nonnesting* permutations, those with no subsequence *abba* with $a \neq b$, introduced by Elizalde [4]. There are exactly $n!\,C_n$ nonnesting permutations of $[n]_2$, where $C_n = \frac{1}{n+1}\binom{2n}{n}$ [4, 5]; we rederive this in Lemma 2.1 below.

Elizalde and Luo [5], extending Luo's thesis [7], carried out a systematic study of pattern avoidance in nonnesting permutations: they enumerated the nonnesting permutations avoiding each set of two or more patterns of length 3, as well as those avoiding several sets of patterns of length 4. (For single patterns of length 3 the noncrossing and nonnesting problems were subsequently advanced by Archer and Laudone [2].) In their concluding section, Elizalde and Luo list a table of conjectured enumeration formulas for five further sets of length-4 patterns [5, Table 4], each checked by computer for $n \leq 8$. The present paper proves the conjecture for the set $\{1132, 3312\}$.

Throughout, a *word* (or *permutation of* $[n]_2$) is a sequence $w = w_1 w_2 \cdots w_{2n}$ over $\{1, \ldots, n\}$ using each letter exactly twice, and containment of a pattern is understood in the sense of [5],

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recalled in Section 2: equalities among pattern letters must be matched by equalities among word letters, and strict inequalities by strict inequalities. A word is *nonnesting* if it avoids 1221 and 2112. Following [5], $\mathcal{C}_n$ denotes the set of nonnesting permutations of $[n]_2$, and $c_n(\Lambda)$ the number of elements of $\mathcal{C}_n$ avoiding every pattern in a set Λ.

Theorem 1.1 (Elizalde–Luo conjecture for $\{1132, 3312\}$)**.** *For every $n \geq 1$,*

$$c_n(\{1132, 3312\}) \;=\; 3^n - 3 \cdot 2^{n-1} + 1.$$

The conjecture appears as the row "$\{1132, 3312\}$ — $3^n - 3 \cdot 2^{n-1} + 1$ — A168583" of [5, Table 4], whose caption reads *"Some conjectures on the enumeration of nonnesting permutations avoiding other patterns"* and whose preamble states that *"All the conjectures have been checked for n up to* 8.*"* The sequence $1, 4, 16, 58, 196, 634, 1996, 6178, \ldots$ given by the formula is the shift $a(n+2)$ of OEIS entry A168583 [8], whose mth term ($m \geq 3$, the entry's own indexing) counts the partitions of the multiset $\{1, 1, 2, 3, \ldots, m-1\}$ into exactly three nonempty parts; as a byproduct, Theorem 1.1 shows that the avoiders of $[n]_2$ are equinumerous with the partitions of $\{1, 1, 2, 3, \ldots, n+1\}$ into exactly three nonempty parts (an equality of cardinalities via the closed form; we exhibit no direct bijection). To our knowledge the conjecture was open before this work: it is stated as a conjecture in the published version of [5] (October 2025) and is unchanged in the latest arXiv revision (v6, January 2026, retrieved June 12, 2026), and we are aware of no announcement of a proof (literature and web searches last performed June 12, 2026).

Outline of the proof

All of the work happens in an explicit, finite combinatorial setting; no generating-function or analytic machinery is needed.

1. **FIFO normal form (Section 2).** A word is nonnesting if and only if, for every i, its ith closer (second occurrence of a value, in position order) carries the same letter as its ith opener. Consequently nonnesting words correspond bijectively to pairs (s, p) where s is a Dyck word of semilength n (the *shape*) and $p \in S_n$ is the sequence of labels read along the openers. We also unwind the patterns 1132 and 3312 into a positional criterion: w contains one of them iff some letter lies strictly between an earlier-closed value v and a letter positioned in between (Corollary 2.4).

2. **Sign words (Section 3).** If (s, p) avoids $\{1132, 3312\}$, the label permutation p must be *prefix-interval*: every prefix of p is an interval of integers. Such p are classically the $\{132, 312\}$-avoiding permutations; they are encoded by a *sign word* $\varepsilon \in \{\mathsf{L}, \mathsf{H}\}^{n-1}$ recording whether each new label is the current minimum or maximum.

3. **The characterization (Section 4).** Theorem 4.1: (s, p) is an avoider iff p is prefix-interval and its sign word satisfies

$$(C): \qquad \varepsilon_K \neq \varepsilon_J \quad \text{for every pair of arcs } K < J \text{ with } q_1 < o_J < q_K,$$

   where o_i, q_i are the positions of the ith opener and closer. The pairs in (C) are crossing pairs of arcs whose later arc opens after the first closer.

4. **Shape classification (Section 5).** The Dyck words s for which (C) is satisfiable form three explicit families (Lemma 5.3), parametrized by the length a of the first ascent, an optional "bridge" position i, and a vector of tail heights $\delta \in \{0, 1\}^{n-a-1}$; on these shapes the valid sign words are exactly the free choices of $|F(s)|$ coordinates, so their number is $2^{|F(s)|}$ (Corollary 5.5).

5. **Two endgames.** Summing $2^{|F(s)|}$ over the three families gives $3^n - 3 \cdot 2^{n-1} + 1$ (Section 6). Independently, Section 7 packages the classification data into a single ternary string and exhibits an explicit bijection, with an explicit inverse, between the avoiders and the language

$$W_n \;=\; \{\mathsf{A},\mathsf{B},\mathsf{C}\}^n \setminus (\mathsf{B}\{\mathsf{A},\mathsf{B}\}^{n-1} \cup \mathsf{C}\{\mathsf{A},\mathsf{B}\}^*\mathsf{C}^*),$$

whose cardinality is $3^n - 2^{n-1} - (2^n - 1)$ by a two-line count. The two endgames are logically independent; either one completes the proof.

Beyond the proofs, every lemma in this paper has been verified mechanically on ranges far exceeding the $n \leq 8$ data that motivated the conjecture, including: an exhaustive word-level verification of Theorem 4.1 for every one of the 2,162,160 nonnesting words at $n = 7$ and every one of the 57,657,600 at $n = 8$; shape-level verification of the classification for all 742,900 Dyck shapes of semilength 13; and an end-to-end check of the bijection on all ground-truth avoiders up to $n = 10$. Appendix A gives the full table of checks and how to reproduce the small-n layer from the literal definitions. Appendix B discloses how this paper was produced (AI-generated mathematics with adversarial machine refereeing) and what was and was not verified by machine.

2 Preliminaries: conventions and the FIFO normal form

2.1 Conventions

We work with the definitions of [5], which we now recall. Let $[n] = \{1, \ldots, n\}$ and let $[n]_2 = \{1, 1, 2, 2, \ldots, n, n\}$ be the multiset with two copies of each element of $[n]$. A *permutation of* $[n]_2$, here called a *word*, is a sequence $w = w_1 \cdots w_{2n}$ over $[n]$ in which every value occurs exactly twice. Positions are 1-based.

Given a word w and a pattern $\sigma = \sigma_1 \cdots \sigma_k$ over the positive integers, w *contains* σ if there exist indices $1 \leq i_1 < \cdots < i_k \leq 2n$ such that for all $r, s \in [k]$,

$$w_{i_r} < w_{i_s} \iff \sigma_r < \sigma_s \qquad \text{and} \qquad w_{i_r} = w_{i_s} \iff \sigma_r = \sigma_s.$$

Otherwise w *avoids* σ. Note both conditions are biconditional: equal pattern letters must map to equal word letters and conversely.[1]

A word is *nonnesting* if it avoids 1221 and 2112. For a value v write $\mathrm{fst}(v) < \mathrm{snd}(v)$ for the positions of its two occurrences; the *arc* of v is the pair $(\mathrm{fst}(v), \mathrm{snd}(v))$. A position is an *opener* if it is the first occurrence of its value, and a *closer* otherwise. Unwinding the containment convention (equal pattern letters force equal values, and each value has exactly two copies), w is nonnesting if and only if there are no two values $u \neq v$ with

$$\mathrm{fst}(u) < \mathrm{fst}(v) < \mathrm{snd}(v) < \mathrm{snd}(u),$$

i.e. no arc nests inside another.

A nonnesting word avoiding both 1132 and 3312 is called an *avoider*, and the number of avoiders of $[n]_2$ is denoted by $c_n = c_n(\{1132, 3312\})$. Avoidance of $\{1132, 3312\}$ is *not* invariant under relabeling of the values, and the count is over all labeled words, with no quotient taken.[2]

We say x is *strictly between* y and z if $y < x < z$ or $z < x < y$. Integer intervals $\{u..v\} = \{j \in \mathbb{Z} : u \leq j \leq v\}$ are understood to be empty when $u > v$.

[1] Before any proof attempt, these conventions were pinned in the project's definitions file as verbatim quotations of the published LaTeX source of [5] (the statement above adapts only the notation), and our computational implementations were validated against the literal definition (Appendix A).

[2] For instance, at $n = 3$ there are 16 avoiders but 3 avoiders whose first occurrences of $1, 2, 3$ appear in increasing order; $3 \cdot 3! = 18 \neq 16$, so the familiar "canonical labeling" shortcut is invalid for this problem.

2.2 Nonnesting words as labeled Dyck shapes

Mark each opener of a word w with U and each closer with D; the resulting word $s = s(w) \in \{\mathsf{U}, \mathsf{D}\}^{2n}$ is the *shape* of w. Since distinct closers in a prefix are preceded by their own distinct openers in the same prefix, every prefix of s has at least as many U's as D's; thus s is a *Dyck word* of semilength n. Write $o_1 < \cdots < o_n$ for the positions of the U's (*openers*) and $q_1 < \cdots < q_n$ for those of the D's (*closers*). In any Dyck word,

$$o_i < q_i \qquad \text{for all } i \in [n], \tag{1}$$

because the prefix ending at q_i contains i letters D, hence at least i letters U.

For a Dyck word s and a permutation $p = (p_1, \dots, p_n)$ of $[n]$, let $\mathrm{wd}(s, p)$ be the word with the letter p_i at positions o_i and q_i for each i.

Lemma 2.1 (FIFO normal form)**.** *A word w over $[n]_2$ is nonnesting if and only if for every $i \in [n]$ the letter at the ith opener of w equals the letter at the ith closer. Consequently, the map $w \mapsto (s(w), p(w))$, where p_i is the letter at the ith opener, is a bijection from the nonnesting words of $[n]_2$ onto the pairs (Dyck word s of semilength n, permutation $p \in S_n$), with inverse $(s, p) \mapsto \mathrm{wd}(s, p)$. In particular $|\mathcal{C}_n| = n!\, C_n$.*

Proof. The arcs of w form a perfect matching of $[2n]$ pairing each opener with a closer; let $\sigma \in S_n$ be the permutation such that the value opened at o_i has its second occurrence at $q_{\sigma(i)}$.

First, $\sigma = \mathrm{id}$ if and only if the letter condition holds. Indeed, if $\sigma = \mathrm{id}$ then the value at q_i is the one opened at o_i. Conversely, suppose $w_{o_i} = w_{q_i} =: v$ for every i. Since v occurs exactly twice, and $o_i \neq q_i$ are an opener and a closer carrying v, the positions o_i, q_i are precisely the two occurrences of v; hence $\sigma(i) = i$.

($\Leftarrow$) Assume $\sigma = \mathrm{id}$ and suppose two values nest: $\mathrm{fst}(u) < \mathrm{fst}(v)$ and $\mathrm{snd}(v) < \mathrm{snd}(u)$. The first inequality says u is opened at an earlier opener, $u = p_i$ and $v = p_j$ with $i < j$; but then $\mathrm{snd}(u) = q_i < q_j = \mathrm{snd}(v)$, a contradiction. So w is nonnesting.

($\Rightarrow$) Assume $\sigma \neq \mathrm{id}$ and pick $i < j$ with $\sigma(i) > \sigma(j)$. The value opened at o_j closes at $q_{\sigma(j)}$, and since the two positions of any one value satisfy fst < snd, we have $o_j < q_{\sigma(j)}$; hence

$$o_i \;<\; o_j \;<\; q_{\sigma(j)} \;<\; q_{\sigma(i)},$$

so the value opened at o_j nests inside the value opened at o_i: w is not nonnesting.

For the bijection: given nonnesting w, the above gives $w = \mathrm{wd}(s(w), p(w))$. Conversely, let s be a Dyck word and $p \in S_n$, and consider $\mathrm{wd}(s, p)$. The position o_i is the first occurrence of p_i: any earlier position carries some p_j with $j < i$ (an earlier U is the jth with $j < i$; an earlier D is the jth D, which by the Dyck property is preceded by at least j U's, so $j \leq i - 1$). Hence the arc of p_i is (o_i, q_i) by (1); arcs sorted simultaneously by opener and closer never nest, so $\mathrm{wd}(s, p)$ is nonnesting, and its shape and label data are (s, p). The two constructions are mutually inverse, and the count $|\mathcal{C}_n| = C_n \cdot n!$ follows. ☐

From now on we identify a nonnesting word with its pair (s, p) and call $i \in [n]$ the *arc* with endpoints (o_i, q_i) and label p_i. For $i < j$, arcs i, j are *crossing* if $o_j < q_i$ and *disjoint* if $q_i < o_j$; by Lemma 2.1 nesting cannot occur.

Remark 2.2 (arc support). Arc i has no letters at positions $> q_i$, and every position $z > q_i$ carries a letter of an arc $J > i$: if $z = o_J$ then $q_J > o_J > q_i$, and if $z = q_J$ then $q_J > q_i$; in either case $q_J > q_i$, which since closers are sorted forces $J > i$.

2.3 Unwinding the patterns 1132 and 3312

Lemma 2.3. *Let w be any word over $[n]_2$. Then:*

1. *w contains 1132 $\iff$ there are a value a and positions $\mathrm{snd}(a) < k < l$ with $a < w_l < w_k$;*
2. *w contains 3312 $\iff$ there are a value c and positions $\mathrm{snd}(c) < k < l$ with $w_k < w_l < c$.*

Proof. (1, $\Rightarrow$) Let $i_1 < i_2 < i_3 < i_4$ be an occurrence of 1132. The pattern letters at i_1, i_2 are equal $(1,1)$, so $w_{i_1} = w_{i_2} =: a$; since a has exactly two copies, $\{i_1, i_2\} = \{\mathrm{fst}(a), \mathrm{snd}(a)\}$ and $i_2 = \mathrm{snd}(a)$. Take $k = i_3$, $l = i_4$. The biconditional convention forces $w_{i_3} > w_{i_4}$ (pattern $3 > 2$) and $w_{i_4} > w_{i_1} = a$ (pattern $2 > 1$), i.e. $a < w_l < w_k$.

(1, $\Leftarrow$) The quadruple of positions $(\mathrm{fst}(a), \mathrm{snd}(a), k, l)$ is increasing, and its letters (a, a, w_k, w_l) realize exactly the order and equality relations of $(1,1,3,2)$: the first two letters are equal and smaller than both others, and $w_k > w_l$, with all stated inequalities strict. So it is an occurrence of 1132.

(2) is the same argument with all inequalities above reversed. □

Corollary 2.4 (combined criterion)**.** *A word w over $[n]_2$ avoids $\{1132, 3312\}$ if and only if there is* no *triple (v, x, y) with v a value and $\mathrm{snd}(v) < x < y \leq 2n$ such that w_y is strictly between v and w_x.*

Proof. "w_y strictly between v and w_x" means $v < w_y < w_x$ (an occurrence of 1132 with $a = v$, by Lemma 2.3(1)) or $w_x < w_y < v$ (an occurrence of 3312 with $c = v$, by Lemma 2.3(2)). □

3 The sign word

For a permutation $p \in S_n$, an occurrence of the (classical) pattern 132 is a triple of indices $i < j < k$ with $p_i < p_k < p_j$, and an occurrence of 312 is $i < j < k$ with $p_j < p_k < p_i$. Having either one means: p_k is strictly between p_i and p_j for some pair $i, j < k$.

Lemma 3.1. *Let $w = (s, p)$ be nonnesting. If p contains 132 or 312, then w contains 1132 or 3312. Hence the label permutation of an avoider avoids $\{132, 312\}$.*

Proof. Suppose $i < j < k$ with p_k strictly between p_i and p_j. The positions $q_i < q_j < q_k$ carry the letters p_i, p_j, p_k, and $q_i = \mathrm{snd}(p_i)$ by Lemma 2.1. Apply Corollary 2.4 with $(v, x, y) = (p_i, q_j, q_k)$: $w_y = p_k$ is strictly between $v = p_i$ and $w_x = p_j$, so w contains a forbidden pattern. □

Write $\mathrm{Av}_n(132, 312)$ for the set of $p \in S_n$ avoiding both patterns.

Lemma 3.2 (structure of $\mathrm{Av}(132, 312)$)**.** *For $p \in S_n$ the following are equivalent:*

1. *p avoids $\{132, 312\}$;*
2. *every p_k is the minimum or the maximum of $\{p_1, \ldots, p_k\}$;*
3. *every prefix set $\{p_1, \ldots, p_k\}$ is an interval of integers ("p is prefix-interval").*

Moreover, the map $p \mapsto \varepsilon(p) = (\varepsilon_2, \ldots, \varepsilon_n) \in \{\mathsf{L}, \mathsf{H}\}^{n-1}$, where $\varepsilon_k = \mathsf{H}$ if $p_k = \max\{p_1, \ldots, p_k\}$ and $\varepsilon_k = \mathsf{L}$ if $p_k = \min\{p_1, \ldots, p_k\}$, is a bijection from $\mathrm{Av}_n(132, 312)$ onto $\{\mathsf{L}, \mathsf{H}\}^{n-1}$, and for all $i < j$,

$$p_i < p_j \iff \varepsilon_j = \mathsf{H} \qquad \text{and} \qquad p_i > p_j \iff \varepsilon_j = \mathsf{L}. \tag{2}$$

We call $\varepsilon(p)$ the sign word *of p, and write $\bar{x}$ for the letter of $\{\mathsf{L}, \mathsf{H}\}$ other than x.*

Proof. (1)⇒(2): if p_k is not extreme in its prefix, there are $u, v < k$ with $p_u < p_k < p_v$; if $u < v$ then (u, v, k) is an occurrence of 132, and if $v < u$ then (v, u, k) is an occurrence of 312.

(2)⇒(3): if some prefix $\{p_1, \ldots, p_k\}$ is not an interval, it omits a value g strictly between its minimum and maximum. Since p is a permutation of $[n]$, $g = p_l$ for some $l > k$; then p_l is neither the minimum nor the maximum of $\{p_1, \ldots, p_l\}$, because the prefix of index k already contains values on both sides of g. This contradicts (2).

(3)⇒(1): under (3), each p_k extends the interval $\{p_1, \ldots, p_{k-1}\}$ by one value at the bottom or the top, hence is never strictly between two earlier values; so no occurrence of 132 or 312 exists.

Bijection: given (3), $\varepsilon(p)$ is well defined (for $k \geq 2$ the prefix has min $<$ max, and p_k is exactly one of min -1, max $+1$). Conversely, given $\varepsilon \in \{\mathsf{L}, \mathsf{H}\}^{n-1}$, set $p_1 = 1 + \#\{j : \varepsilon_j = \mathsf{L}\}$ and iteratively $p_k = (\text{current min}) - 1$ or $(\text{current max}) + 1$ according to ε_k. This is the unique preimage and lands in S_n: the final prefix is an interval of length n equal to $[\, p_1 - \#\mathsf{L},\ p_1 + \#\mathsf{H}\,] = [1, n]$. The two maps are mutually inverse by induction on k.

(2): if $\varepsilon_j = \mathsf{H}$ then $p_j = \max\{p_1, \ldots, p_j\} > p_i$ for every $i < j$; if $\varepsilon_j = \mathsf{L}$ then $p_j < p_i$. The two cases are exhaustive and exclusive. □

Remark 3.3. The equivalence of (1)–(3) and the count $|\operatorname{Av}_n(132, 312)| = 2^{n-1}$ are classical [9]; we have included the short proof to keep the paper self-contained and because the reconstruction map and rule (2) are used constantly below.

4 The characterization

Theorem 4.1. *Let $w = (s, p)$ be a nonnesting word with $p \in \operatorname{Av}_n(132, 312)$ and sign word ε. Then w avoids $\{1132, 3312\}$ if and only if*

$$(C): \qquad \varepsilon_K \neq \varepsilon_J \quad \textit{for every pair } K < J \textit{ with } q_1 < o_J < q_K.$$

Note that the pairs constrained by (C) are crossing pairs of arcs ($K < J$ and $o_J < q_K$) whose later arc opens after the *first* closer of the word; automatically $K \geq 2$, since $K = 1$ would require $o_J < q_1$.

Proof. By Corollary 2.4, w contains a forbidden pattern iff there is a triple (v, x, y) with $v = p_i$ for some arc i (every value is an arc label, and $\operatorname{snd}(p_i) = q_i$ by Lemma 2.1), positions $q_i < x < y$, and w_y strictly between p_i and w_x. By Remark 2.2, x and y are letters of arcs $J, K > i$ respectively; and $J \neq K$, since strict betweenness forces $w_x \neq w_y$.

Case $K > J$. Then $i, J < K$, so $p_i, p_J \in \{p_1, \ldots, p_K\}$, and by Lemma 3.2(2) the letter $w_y = p_K$ is the minimum or maximum of that set — never strictly between p_i and p_J. So no violating triple has $K > J$.

Case $K < J$ (hence $i < K < J$). Arc K's positions satisfy $o_K < o_J$ and $q_K < q_J$. Since x is a letter of J and $y > x$ is a letter of K: if $x = q_J$, then $y > q_J$, but both letters of arc K sit at positions $< q_J$ — impossible; so $x = o_J$. If $y = o_K$, then $y = o_K < o_J = x$ — impossible; so $y = q_K$. The requirement $x < y$ reads $o_J < q_K$, and $x > q_i$ reads $q_i < o_J$. As for betweenness, rule (2) gives: if $\varepsilon_J = \mathsf{H}$ then $p_J > p_i$ and $p_J > p_K$, so p_K is strictly between p_i and p_J iff $p_i < p_K$ iff $\varepsilon_K = \mathsf{H}$; if $\varepsilon_J = \mathsf{L}$ then p_J lies below both, and betweenness holds iff $p_K < p_i$ iff $\varepsilon_K = \mathsf{L}$. In both cases:

$$p_K \text{ strictly between } p_i \text{ and } p_J \iff \varepsilon_K = \varepsilon_J.$$

Crucially, this criterion does not depend on the witness arc i: rule (2) gives $p_i < p_K \iff \varepsilon_K = \mathsf{H}$ *for every $i < K$*.

Summarizing, a violating triple exists iff there are arcs $i < K < J$ with $q_i < o_J < q_K$ and $\varepsilon_K = \varepsilon_J$. Finally, fix $K < J$ with $o_J < q_K$; we claim an arc $i < K$ with $q_i < o_J$ exists iff $q_1 < o_J$. Indeed, if $q_1 < o_J$ then $i = 1$ works ($1 < K$ holds because $q_1 < o_J < q_K$ forces $K \neq 1$); conversely any such i has $q_1 \leq q_i < o_J$. Hence w contains a forbidden pattern iff some pair $K < J$ with $q_1 < o_J < q_K$ has $\varepsilon_K = \varepsilon_J$, i.e. iff (C) fails. ☐

Corollary 4.2. *The map $w \mapsto (s(w), \varepsilon(p(w)))$ is a bijection from the set of avoiders of $[n]_2$ onto the set of pairs (s, ε) where s is a Dyck word of semilength n and $\varepsilon \in \{\mathsf{L}, \mathsf{H}\}^{n-1}$ satisfies (C) for s. We call such ε* valid *for s.*

Proof. Combine Lemma 2.1 ($w \leftrightarrow (s, p)$, a bijection), Lemma 3.1 (avoiders have $p \in \mathrm{Av}(132, 312)$, so none are lost in passing to sign words), Lemma 3.2 ($p \leftrightarrow \varepsilon$, a bijection), and Theorem 4.1 (avoidance $\iff$ (C)). For $n = 1$, ε is the empty word, (C) is vacuous, and the unique word 11 is an avoider. ☐

Example 4.3. Let $n = 3$ and $s = \mathsf{UUDUDD}$, so $(o_1, o_2, o_3) = (1, 2, 4)$ and $(q_1, q_2, q_3) = (3, 5, 6)$. The only pair (K, J) with $q_1 < o_J < q_K$ is $(2, 3)$: $q_1 = 3 < o_3 = 4 < q_2 = 5$. So (C) reads $\varepsilon_2 \neq \varepsilon_3$, with the two solutions $\varepsilon \in \{\mathsf{LH}, \mathsf{HL}\}$, i.e. $p \in \{(2, 1, 3), (2, 3, 1)\}$. The word pattern of this shape is $p_1p_2p_1p_3p_2p_3$, so the two avoiders are 212313 and 232131. Indeed the other four label permutations all fail; e.g. $p = (1, 2, 3)$ gives 121323, which contains 1132 as the subsequence 1 1 3 2 at positions $1, 3, 4, 5$. This matches Theorem 4.1, and the count 2 will reappear in Example 7.6 as the two codes $\mathsf{CCA}, \mathsf{CCB}$.

5 The admissible shapes

Fix a Dyck word s of semilength n. Let $a = a(s) \geq 1$ be the length of its first ascent run, so that $o_j = j$ for $j \leq a$ and $q_1 = a + 1$. Call an arc J *late* if $o_J > q_1$. For a late arc J let

$$S_J \ = \ \{K < J : o_J < q_K\},$$

so that the pairs constrained by (C) are exactly the pairs (K, J) with J late and $K \in S_J$. (For $K < J$ we have $o_K < o_J$ automatically, so S_J is the set of arcs *open* at the moment o_J, i.e. with $o_K < o_J < q_K$.)

Lemma 5.1. *Let $J \in [n]$ and let c be the number of closers at positions $< o_J$. Then $S_J = \{c+1, \dots, J-1\}$, and $|S_J| = J - 1 - c$ equals the height of s (number of U's minus number of D's) just before position o_J. Moreover, J is late if and only if $J \geq a + 1$.*

Proof. The arcs opened before o_J are exactly $1, \dots, J - 1$ (openers are sorted). The closers before o_J are an initial segment of $q_1 < q_2 < \cdots$, hence are exactly $q_1, \dots, q_c$: arcs $1, \dots, c$ are closed before o_J and arcs $c + 1, \dots, J - 1$ remain open ($c \leq J - 1$ since each closed arc was opened). The height before a position equals the number of earlier openers minus earlier closers, i.e. $(J - 1) - c$. Finally, openers $1, \dots, a$ sit at positions $1, \dots, a < q_1 = a + 1$, and every opener o_J with $J \geq a + 1$ satisfies $o_J > a + 1 = q_1$ because position $a + 1$ is a closer. ☐

Define, with respect to the decomposition above:

- a *gap opener*: an opener at a position strictly between q_1 and q_a (for $a = 1$ there is none, as the interval is empty);
- a *tail opener*: an opener at a position $> q_a$; the corresponding arcs are *tail arcs*. For a tail arc j, let $\delta_j \in \{0, 1, 2, \dots\}$ be the height of s just before position o_j (its *start height*).

Lemma 5.2 (obstructions)**.** *If s has at least two gap openers, or a tail opener with start height ≥ 2, then no $\varepsilon \in \{\mathsf{L}, \mathsf{H}\}^{n-1}$ is valid for s.*

Proof. Two gap openers. Then $a \geq 2$. The openers in the interval (q_1, q_a) belong to arcs $a+1, a+2, \dots$ in order (openers are sorted and arcs $\leq a$ open before q_1); let $\beta = a+1$ and $\beta' = a+2$ be the arcs of the first two. Both are late. Let $c \leq c'$ be the numbers of closers before o_β and $o_{\beta'}$; since $q_1 < o_\beta < o_{\beta'} < q_a$ we get $1 \leq c \leq c' \leq a-1$. By Lemma 5.1, $S_\beta = \{c+1, \dots, a\} \ni a$ and $S_{\beta'} = \{c'+1, \dots, a+1\} \supseteq \{a, \beta\}$. So the pairs (a, β), (a, β'), (β, β') are all constrained by (C), and a valid ε would need $\varepsilon_a, \varepsilon_\beta, \varepsilon_{\beta'}$ pairwise distinct in the two-element set $\{\mathsf{L}, \mathsf{H}\}$ — impossible.

Tail opener at height ≥ 2. Let J be its arc, so $o_J > q_a$ and, by Lemma 5.1, $S_J = \{c+1, \dots, J-1\}$ with $J-1-c \geq 2$; hence $K := J-2$ and $K' := J-1$ both lie in S_J. Note $K \geq c+1 \geq a+1 \geq 2$, since at least the a closers $q_1, \dots, q_a$ precede o_J. The arc K' is late: $q_{K'} > o_J > q_a$ forces $K' > a$ (closers are sorted), so $K' \geq a+1$. Also $K \in S_{K'}$: $K < K'$ and $q_K > o_J > o_{K'}$. So the pairs (K, K'), (K, J), (K', J) are all constrained by (C), again requiring three pairwise distinct values in $\{\mathsf{L}, \mathsf{H}\}$ — impossible. ☐

It remains to describe the shapes with at most one gap opener and all tail start heights ≤ 1; we call these *admissible*. For $m \geq 0$, $h_0 \in \{0, 1\}$ and $d_1, \dots, d_m \in \{0, 1\}$ with $d_1 \leq h_0$ when $m \geq 1$, let $\mathrm{Tail}(h_0; d_1, \dots, d_m)$ be the $\{\mathsf{U}, \mathsf{D}\}$-word produced by the following scan: start with $h := h_0$; for $r = 1, \dots, m$ emit D^{h-d_r} then U, and set $h := d_r + 1$; finally emit D^{h}. (All D-run lengths are ≥ 0: $d_1 \leq h_0$, and for $r \geq 2$, $d_r \leq 1 \leq d_{r-1} + 1 = h$. For $m = 0$ the word is D^{h_0}.) By construction, Tail describes a lattice path from height h_0 to height 0, never negative, whose rth U-step starts at height d_r.

Lemma 5.3 (classification)**.** *Let s be admissible. Then exactly one of the following holds, in each case with the stated parameters determined by s; and conversely, every choice of parameters yields an admissible Dyck word of semilength n with those data.*

(S) Staircase*: $a = n$ and $s = \mathsf{U}^n \mathsf{D}^n$. (No parameters.)*

(I) $1 \leq a \leq n-1$ and s has no gap opener. The tail arcs are $a+1, \dots, n$; $\delta_{a+1} = 0$; and

$$s = \mathsf{U}^a \mathsf{D}^a \cdot \mathrm{Tail}(0;\, \delta_{a+1}, \dots, \delta_n),$$

with parameters $\big(a,\ \delta = (\delta_{a+2}, \dots, \delta_n) \in \{0,1\}^{n-a-1}\big)$ and $\delta_{a+1} := 0$.

(II) $2 \leq a \leq n-1$ and s has exactly one gap opener, which belongs to the arc $B := a+1$ and has exactly i closers before it, where $1 \leq i \leq a-1$; moreover $S_B = \{i+1, \dots, a\}$. The tail arcs are $a+2, \dots, n$, and

$$s = \mathsf{U}^a \mathsf{D}^{\,i}\, \mathsf{U}\, \mathsf{D}^{\,a-i} \cdot \mathrm{Tail}(1;\, \delta_{a+2}, \dots, \delta_n),$$

with parameters $\big(a,\ i,\ \delta = (\delta_{a+2}, \dots, \delta_n) \in \{0,1\}^{n-a-1}\big)$.

Proof. If $a = n$ then all n arcs open in the first run and $s = \mathsf{U}^n \mathsf{D}^n$: case (S). Assume $a < n$.

No gap opener. Every position strictly between q_1 and q_a is a closer, so $q_m = a + m$ for $m \leq a$ and the word begins $\mathsf{U}^a \mathsf{D}^a$. All arcs $\geq a+1$ open after $q_a = 2a$: the tail arcs are exactly $a+1, \dots, n$, a nonempty set since $a < n$. The height after position $2a$ is 0 and the next position must be an opener (a closer would make the height negative), so $\delta_{a+1} = 0$; by admissibility $\delta_j \leq 1$ for the rest. The remainder of the word is forced by the start heights: between consecutive tail openers all positions are closers, and the height must drop from $\delta_j + 1$ (just after the U of tail arc j) to δ_{j+1} (just before the U of tail arc $j+1$), giving exactly $\delta_j + 1 - \delta_{j+1}$ letters D; after the last opener the height $\delta_n + 1$ returns to 0. This is precisely $\mathrm{Tail}(0; \delta_{a+1}, \ldots, \delta_n)$, so s has the displayed form and (a, δ) is determined by s. Conversely, for any (a, δ) the displayed word is a Dyck word of semilength n (U-count $a + (n - a) = n$; the scan keeps $h \geq 0$ and ends at height 0), its first ascent is exactly a (position $a + 1$ is a D since $a \geq 1$), it has no gap opener (positions $a + 1, \ldots, 2a$ are D's, and $q_a = 2a$), and its tail start heights are $\delta_{a+1}, \ldots, \delta_n$ by the construction of the scan: case (I) with the same parameters.

One gap opener. The gap opener is the first opener after q_1, hence belongs to arc $a+1 =: B$; the interval (q_1, q_a) must be nonempty, so $a \geq 2$. Let i be the number of closers before o_B: $o_B > q_1$ gives $i \geq 1$ and $o_B < q_a$ gives $i \leq a - 1$. Positions in (q_1, q_a) other than o_B are closers, so the prefix of s up to q_a reads $\mathsf{U}^a\mathsf{D}^i\mathsf{U}\mathsf{D}^{a-i}$ (closers $q_1, \ldots, q_i$, then o_B, then $q_{i+1}, \ldots, q_a$); in particular $q_a = 2a + 1$ and, by Lemma 5.1 with $c = i$, $S_B = \{i + 1, \ldots, a\}$. The height after q_a is $(a + 1) - a = 1$. The tail arcs are $a + 2, \ldots, n$ (possibly none), and the same forced-run argument as in case (I) identifies the remainder with $\mathrm{Tail}(1; \delta_{a+2}, \ldots, \delta_n)$; for $a = n - 1$ this is $\mathrm{Tail}(1; \,) = \mathsf{D}$. So (a, i, δ) is determined by s, and conversely each (a, i, δ) yields a Dyck word of case (II) by the same checks (U-count $a + 1 + (n - a - 1) = n$; first ascent a since $i \geq 1$ puts a D at position $a + 1$; exactly one gap opener, at position $a + 1 + i < 2a + 1 = q_a$, with i closers before it; tail start heights δ_j).

The three cases are pairwise exclusive ($a = n$ versus $a < n$; gap opener absent versus present), and every admissible shape falls in one of them. □

Lemma 5.4 (local form of validity)**.** *Let s be admissible and $\varepsilon \in \{\mathsf{L}, \mathsf{H}\}^{n-1}$ (indexed $2, \ldots, n$). Then ε is valid for s if and only if:*

- *(S) always (there are no late arcs, hence no constrained pairs);*
- *(I) for every tail arc $j \geq a + 2$ with $\delta_j = 1$: $\varepsilon_j = \overline{\varepsilon_{j-1}}$;*
- *(II) $\varepsilon_{i+1} = \cdots = \varepsilon_a = \overline{\varepsilon_{a+1}}$, and for every tail arc $j \geq a + 2$ with $\delta_j = 1$: $\varepsilon_j = \overline{\varepsilon_{j-1}}$.*

Proof. Validity is the conjunction, over late arcs J (i.e. $J \geq a + 1$, by Lemma 5.1) and $K \in S_J$, of $\varepsilon_K \neq \varepsilon_J$. We list the sets S_J case by case using Lemma 5.1.

(S): $a = n$ and there are no late arcs; the conjunction is empty.

(I): the late arcs are $a + 1, \ldots, n$, all tail arcs. For $J = a + 1$: the closers before $o_{a+1} = 2a + 1$ are $q_1, \ldots, q_a$, so $S_{a+1} = \{a+1, \ldots, a\} = \varnothing$ — no constraint. For a tail arc $J \geq a+2$: $|S_J| = \delta_J \in \{0, 1\}$ by Lemma 5.1. If $\delta_J = 0$ there is no constraint; if $\delta_J = 1$ then S_J, being the interval $\{c+1, \ldots, J-1\}$ of size one, equals $\{J - 1\}$, and the constraint $\varepsilon_J \neq \varepsilon_{J-1}$ over a two-letter alphabet reads $\varepsilon_J = \overline{\varepsilon_{J-1}}$.

(II): the late arcs are $a + 1, \ldots, n$. For $J = B = a + 1$: $S_B = \{i + 1, \ldots, a\}$ by Lemma 5.3, and the constraints $\{\varepsilon_K \neq \varepsilon_B : K = i + 1, \ldots, a\}$ hold iff $\varepsilon_{i+1} = \cdots = \varepsilon_a = \overline{\varepsilon_{a+1}}$ (the block is nonempty as $i \leq a - 1$). For tail arcs $J \geq a + 2$ the analysis is exactly as in (I); note that for $J = a + 2$ with $\delta_J = 1$ the constraint set is $S_J = \{a + 1\} = \{B\}$, also covered by the rule $\varepsilon_J = \overline{\varepsilon_{J-1}}$. □

Corollary 5.5 (free coordinates)**.** *For each admissible s define the* free set *$F(s) \subseteq \{2, \ldots, n\}$:*

$$F(s) = \begin{cases} \{2..n\} & \textit{(S)}, \\ \{2..a + 1\} \cup \{j \geq a + 2 : \delta_j = 0\} & \textit{(I)}, \\ \{2..i\} \cup \{a + 1\} \cup \{j \geq a + 2 : \delta_j = 0\} & \textit{(II)}. \end{cases}$$

Then the restriction map $\varepsilon \mapsto \varepsilon|_{F(s)}$ is a bijection from the valid sign words for s onto $\{\mathsf{L}, \mathsf{H}\}^{F(s)}$.

In particular the number of valid sign words is $N(s) = 2^{|F(s)|}$, *with*

$$|F(s)| = \begin{cases} n-1 & \textit{(S)}, \\ a+z & \textit{(I)}, \\ i+z & \textit{(II)}, \end{cases} \qquad \textit{where } z := \#\{a+2 \le j \le n : \delta_j = 0\}.$$

Proof. Define an extension map E on $f \in \{\mathsf{L}, \mathsf{H}\}^{F(s)}$: set $\varepsilon_j := f_j$ for $j \in F(s)$; in case (II) set $\varepsilon_{i+1} = \cdots = \varepsilon_a := \overline{f_{a+1}}$; then process the tail arcs $j = a+2, \ldots, n$ in increasing order, setting $\varepsilon_j := \overline{\varepsilon_{j-1}}$ whenever $\delta_j = 1$. Each reference ε_{j-1} is already defined at the time it is used: $j-1$ is in $F(s)$, or in the block $\{i+1..a\}$, or equals $a+1$, or is an earlier-processed tail arc. By construction $E(f)$ satisfies the conditions of Lemma 5.4, hence is valid, and $E(f)|_{F(s)} = f$. Conversely, if ε is valid, then by Lemma 5.4 its non-free coordinates are determined from $\varepsilon|_{F(s)}$ by exactly the assignments that E makes — the block coordinates equal $\overline{\varepsilon_{a+1}}$, and each $\delta_j = 1$ tail coordinate equals $\overline{\varepsilon_{j-1}}$, by induction on j — so $E(\varepsilon|_{F(s)}) = \varepsilon$. Thus restriction and extension are mutually inverse. The values of $|F(s)|$ are immediate counts: in case (S), $|\{2..n\}| = n-1$; in case (I), $|\{2..a+1\}| = a$, giving $a+z$; in case (II), $|\{2..i\}| = i-1$ plus the singleton $\{a+1\}$, giving $i+z$. $\square$

6 First proof of the main theorem: summation

Theorem 6.1. *For every* $n \ge 1$, *the number of nonnesting permutations of* $[n]_2$ *avoiding* 1132 *and* 3312 *is*

$$c_n \;=\; 3^n - 3 \cdot 2^{n-1} + 1.$$

Proof. By Corollary 4.2, $c_n = \sum_s N(s)$, the sum over all Dyck words s of semilength n of the number of valid sign words. By Lemma 5.2, $N(s) = 0$ unless s is admissible, and by Lemma 5.3 the admissible shapes are parametrized bijectively by the data (S), (a, δ) in case (I) with $1 \le a \le n-1$, and (a, i, δ) in case (II) with $2 \le a \le n-1$, $1 \le i \le a-1$, where $\delta \in \{0,1\}^{n-a-1}$ in both cases. By Corollary 5.5, $N(s) = 2^{|F(s)|}$ with the listed exponents. We sum case by case. Since each coordinate δ_j ranges freely over $\{0,1\}$ and contributes a factor 2 to $2^{|F(s)|}$ when $\delta_j = 0$ and 1 when $\delta_j = 1$,

$$\sum_{\delta \in \{0,1\}^{n-a-1}} 2^{\,z(\delta)} \;=\; (2+1)^{\,n-a-1} \;=\; 3^{\,n-a-1}.$$

Hence

$$c_n \;=\; \underbrace{2^{n-1}}_{\text{(S)}} + \underbrace{\sum_{a=1}^{n-1} 2^a\, 3^{\,n-1-a}}_{\text{(I)}} + \underbrace{\sum_{a=2}^{n-1}\Big(\sum_{i=1}^{a-1} 2^i\Big) 3^{\,n-1-a}}_{\text{(II)}} \;=\; 2^{n-1} \;+\; \Sigma_1 \;+\; \Sigma_2,$$

with $\sum_{i=1}^{a-1} 2^i = 2^a - 2$. We use the identity

$$\sum_{a=1}^{m} 2^{a-1} 3^{\,m-a} \;=\; 3^m - 2^m \qquad (m \ge 0), \tag{3}$$

proved by induction on m: both sides vanish at $m = 0$ and satisfy $g(m) = 3g(m-1) + 2^{m-1}$. Applying (3) with $m = n-1$,

$$\Sigma_1 = 2 \sum_{a=1}^{n-1} 2^{a-1} 3^{(n-1)-a} = 2\big(3^{n-1} - 2^{n-1}\big).$$

For Σ_2, assume first that $n \geq 3$. Removing the $a = 1$ term $2 \cdot 3^{n-2}$ from Σ_1 and using $\sum_{a=2}^{n-1} 3^{n-1-a} = \sum_{k=0}^{n-3} 3^k = \frac{3^{n-2}-1}{2}$,

$$\Sigma_2 = \Big(2\big(3^{n-1} - 2^{n-1}\big) - 2 \cdot 3^{n-2}\Big) - 2 \cdot \frac{3^{n-2}-1}{2} = 2 \cdot 3^{n-1} - 3 \cdot 3^{n-2} - 2^n + 1 = 3^{n-1} - 2^n + 1.$$

For $n \in \{1, 2\}$ the sum Σ_2 is empty and the expression $3^{n-1} - 2^n + 1$ equals 0, so this closed form holds for all $n \geq 1$. Adding up,

$$c_n = 2^{n-1} + 2 \cdot 3^{n-1} - 2^n + 3^{n-1} - 2^n + 1 = 3^n + 2^{n-1} - 2^{n+1} + 1 = 3^n - 3 \cdot 2^{n-1} + 1. \qquad \square$$

Remark 6.2. The three exponential scales of the answer are transparent in the decomposition: each free tail coordinate contributes a factor 3 (the choices "$\delta_j = 0$ with $\varepsilon_j = \mathsf{L}$", "$\delta_j = 0$ with $\varepsilon_j = \mathsf{H}$", "$\delta_j = 1$, sign forced"), the first-run coordinates contribute factors of 2, and the bookkeeping of the three families produces exactly $3^n - 3 \cdot 2^{n-1} + 1$. This matches the ordinary generating function $x(1 - 2x + 3x^2)/((1 - x)(1 - 2x)(1 - 3x))$ of the (shifted) OEIS entry A168583 [8], with poles at $1, \frac{1}{2}, \frac{1}{3}$.

7 Second proof of the endgame: an explicit bijection

The summation in Section 6 can be replaced by an explicit bijection: the classification data $(a, i, \delta, \varepsilon|_{F(s)})$ of an avoider can be packaged into a single ternary string, in such a way that the image language has a transparent description and count. Throughout this section, fix the alphabet $\{\mathsf{A}, \mathsf{B}, \mathsf{C}\}$ and identify signs with letters via

$$\ell(\mathsf{L}) = \mathsf{A}, \qquad \ell(\mathsf{H}) = \mathsf{B}.$$

Definition 7.1. $W_n \subseteq \{\mathsf{A}, \mathsf{B}, \mathsf{C}\}^n$ is the set of strings σ such that:

- if σ contains no C, then $\sigma_1 = \mathsf{A}$;
- if $\sigma_1 = \mathsf{C}$, then there are positions $2 \leq r < t \leq n$ with $\sigma_r = \mathsf{C}$ and $\sigma_t \neq \mathsf{C}$;
- (if $\sigma_1 \neq \mathsf{C}$ and σ contains a C: no condition).

Lemma 7.2. *The complement of W_n in $\{\mathsf{A}, \mathsf{B}, \mathsf{C}\}^n$ is the disjoint union of*

$$X_1 = \mathsf{B}\,\{\mathsf{A}, \mathsf{B}\}^{n-1} \quad \textit{and} \quad X_2 = \{\, \mathsf{C}\, v\, \mathsf{C}^k \ :\ v \in \{\mathsf{A}, \mathsf{B}\}^m,\ m, k \geq 0,\ 1 + m + k = n \,\},$$

with $|X_1| = 2^{n-1}$ and $|X_2| = \sum_{m=0}^{n-1} 2^m = 2^n - 1$. Hence

$$|W_n| \ =\ 3^n - 2^{n-1} - (2^n - 1) \ =\ 3^n - 3 \cdot 2^{n-1} + 1 \qquad (n \geq 1).$$

Proof. A string violates Definition 7.1 iff either it is C-free with $\sigma_1 = \mathsf{B}$ (a C-free string cannot have $\sigma_1 = \mathsf{C}$), or $\sigma_1 = \mathsf{C}$ and no C at a position $r \geq 2$ is followed later by a non-C. The first family is X_1. For the second, write $\sigma = \mathsf{C}\rho$: the condition says that every letter after the first C of ρ (if any) is again C, i.e. $\rho \in \{\mathsf{A}, \mathsf{B}\}^* \mathsf{C}^*$, which is exactly the family X_2 (the case $k = 0$ covering C-free ρ). X_1 and X_2 are disjoint (first letter B versus C), and within X_2 the pair (m, k) is determined by σ (v is the maximal $\{\mathsf{A}, \mathsf{B}\}$-prefix of ρ), so $|X_2| = \sum_{m+k=n-1} 2^m = 2^n - 1$. $\square$

Definition 7.3 (the encoding Φ)**.** Let s be admissible and ε valid for s. Define $\sigma = \Phi(s, \varepsilon) \in \{\mathsf{A}, \mathsf{B}, \mathsf{C}\}^n$ by cases on Lemma 5.3:

(S) $\sigma_1 = \mathsf{A}$; and $\sigma_j = \ell(\varepsilon_j)$ for $2 \le j \le n$.

(I) $\sigma_j = \ell(\varepsilon_{j+1})$ for $1 \le j \le a$; $\sigma_{a+1} = \mathsf{C}$; and for $a+2 \le j \le n$: $\sigma_j = \mathsf{C}$ if $\delta_j = 1$, else $\sigma_j = \ell(\varepsilon_j)$.

(II) $\sigma_1 = \mathsf{C}$; $\sigma_j = \ell(\varepsilon_j)$ for $2 \le j \le i$; $\sigma_j = \mathsf{C}$ for $i+1 \le j \le a$; $\sigma_{a+1} = \ell(\varepsilon_{a+1})$; and for $a+2 \le j \le n$: $\sigma_j = \mathsf{C}$ if $\delta_j = 1$, else $\sigma_j = \ell(\varepsilon_j)$.

In words: every free coordinate of ε is written as its sign letter, every *forced* arc (the block $\{i+1..a\}$ of case (II), and the $\delta_j = 1$ tail arcs) is written C; in case (I) the initial free block $\varepsilon_2, \dots, \varepsilon_{a+1}$ is shifted one slot to the left so that position $a+1$ can carry a C marking the end of the first descent; in case (II), position 1 carries a C marking the presence of the gap opener; in case (S), position 1 carries the padding letter A.

Theorem 7.4. *Φ is a bijection from $\{(s,\varepsilon) : s$ admissible, ε valid for $s\}$ onto W_n.*

Proof. **Step 1: Φ lands in W_n, with disjoint images for the three cases.** (S): σ is C-free with $\sigma_1 = \mathsf{A}$, so $\sigma \in W_n$. (I): $\sigma_1 = \ell(\varepsilon_2) \in \{\mathsf{A}, \mathsf{B}\}$ and σ contains a C (at position $a+1 \le n$), so $\sigma \in W_n$ by the third clause. (II): $\sigma_1 = \mathsf{C}$; take $r = i+1$ and $t = a+1$: then $2 \le r < t \le n$, $\sigma_r = \mathsf{C}$ (the block is nonempty since $i \le a-1$) and $\sigma_t \in \{\mathsf{A}, \mathsf{B}\}$, so $\sigma \in W_n$ by the second clause. Disjointness: (S)-images are C-free; (I)-images contain a C and have $\sigma_1 \ne \mathsf{C}$; (II)-images have $\sigma_1 = \mathsf{C}$.

Step 2: a parse map Ψ defined on all of W_n. Let $\sigma \in W_n$.

Case $\mathsf{C} \notin \sigma$. Then $\sigma_1 = \mathsf{A}$ by Definition 7.1. Output case (S) with $\varepsilon_j := \ell^{-1}(\sigma_j)$ for $2 \le j \le n$.

Case $\sigma_1 \ne \mathsf{C}$, $\mathsf{C} \in \sigma$. Let $a+1$ be the position of the first C, so $1 \le a \le n-1$ and $\sigma_1, \dots, \sigma_a \in \{\mathsf{A}, \mathsf{B}\}$. Set $\varepsilon_{j+1} := \ell^{-1}(\sigma_j)$ for $1 \le j \le a$; then for $j = a+2, \dots, n$ in increasing order: if $\sigma_j = \mathsf{C}$ set $\delta_j := 1$ and $\varepsilon_j := \overline{\varepsilon_{j-1}}$, else set $\delta_j := 0$ and $\varepsilon_j := \ell^{-1}(\sigma_j)$. Output case (I) with parameters (a, δ) — an admissible shape by the converse direction of Lemma 5.3(I) — and the sign word ε, which is valid by Lemma 5.4(I), since its $\delta_j = 1$ coordinates satisfy the forced equalities by construction.

Case $\sigma_1 = \mathsf{C}$. Let $i - 1 \ge 0$ be the length of the maximal $\{\mathsf{A}, \mathsf{B}\}$-run starting at position 2, so positions $2, \dots, i$ carry letters in $\{\mathsf{A}, \mathsf{B}\}$ and position $i+1$, if it exists, carries C. The second clause of Definition 7.1 provides $2 \le r < t \le n$ with $\sigma_r = \mathsf{C}$ and $\sigma_t \ne \mathsf{C}$; in particular there is a C at a position ≥ 2, so $i+1 \le n$ and $\sigma_{i+1} = \mathsf{C}$. Let the maximal C-run starting at position $i+1$ occupy positions $i+1, \dots, a$. We claim $a \le n-1$ and $\sigma_{a+1} \in \{\mathsf{A}, \mathsf{B}\}$. Consider the witness pair (r,t), and note $r \ge i+1$ (positions $2, \dots, i$ are not C). If $r \le a$: then $t > r$ and $\sigma_t \ne \mathsf{C}$ force $t \ge a+1$ (positions $r+1, \dots, a$ are C), so position $a+1$ exists, and $\sigma_{a+1} \ne \mathsf{C}$ by maximality of the run. If $r > a$: then $a+1 \le r \le n$, so position $a+1$ exists; $r = a+1$ is impossible (a C at $a+1$ would extend the maximal run), so again $\sigma_{a+1} \ne \mathsf{C}$ by maximality. This proves the claim. Now set $\varepsilon_j := \ell^{-1}(\sigma_j)$ for $2 \le j \le i$; $\varepsilon_{a+1} := \ell^{-1}(\sigma_{a+1})$; $\varepsilon_{i+1} = \cdots = \varepsilon_a := \overline{\varepsilon_{a+1}}$; and process $j = a+2, \dots, n$ exactly as in the previous case (defining δ_j and ε_j). Output case (II) with parameters (a, i, δ) — well defined, since $1 \le i \le a-1 \le n-2$, and an admissible shape by Lemma 5.3(II) — and the sign word ε, valid by Lemma 5.4(II).

Step 3: $\Psi \circ \Phi = \mathrm{id}$. Let (s, ε) be a valid pair and $\sigma = \Phi(s, \varepsilon)$.

- (S): σ is C-free, so Ψ takes the (S)-branch and recovers each ε_j from σ_j.
- (I): $\sigma_1 \ne \mathsf{C}$, and the first C of σ is at position $a+1$, since positions $1, \dots, a$ carry sign letters. So Ψ takes the (I)-branch and recovers a; it recovers $\varepsilon_2, \dots, \varepsilon_{a+1}$ from positions $1, \dots, a$; and for $j \ge a+2$ it recovers δ_j ($\mathsf{C} \leftrightarrow \delta_j = 1$) and ε_j — directly if $\delta_j = 0$, and as $\overline{\varepsilon_{j-1}}$ if $\delta_j = 1$, which equals the original ε_j because the original is valid (Lemma 5.4(I)); the argument is an

induction on j using that ε_{j-1} was recovered correctly. By Lemma 5.3(I), the parameters (a, δ) determine s.

- (II): $\sigma_1 = \mathsf{C}$, so Ψ takes the (II)-branch. In σ, positions $2, \ldots, i$ carry sign letters, positions $i+1, \ldots, a$ carry C (a nonempty block), and position $a+1$ carries a sign letter; hence the maximal $\{\mathsf{A}, \mathsf{B}\}$-run from position 2 ends exactly at i and the maximal C-run from $i+1$ ends exactly at a: Ψ recovers (i, a). It then recovers $\varepsilon_2, \ldots, \varepsilon_i$ and ε_{a+1} directly; the block coordinates as $\overline{\varepsilon_{a+1}}$ — correct by Lemma 5.4(II) — and the tail as in case (I). By Lemma 5.3(II), (a, i, δ) determine s.

Step 4: $\Phi \circ \Psi = \mathrm{id}$**.** Let $\sigma \in W_n$ and $(s, \varepsilon) = \Psi(\sigma)$. In each branch, the output of Ψ classifies into the same case it was parsed in (this is the converse direction of Lemma 5.3, which asserts that the constructed shape lies in case (I), resp. (II), with the same parameters), and Φ applied to it writes back exactly the letters read: in (S), the letter A followed by the sign letters; in (I), the a sign letters read from positions $1, \ldots, a$, the C at position $a+1$, and for $j \geq a+2$ a C iff $\delta_j = 1$ iff $\sigma_j = \mathsf{C}$, else $\ell(\varepsilon_j) = \sigma_j$; in (II), the leading C, the run structure (positions $2, \ldots, i$; the block $i+1, \ldots, a$; position $a+1$), and the tail, likewise. So $\Phi(\Psi(\sigma)) = \sigma$.

By Steps 1–4, Φ is a bijection with inverse Ψ. □

Corollary 7.5 (second proof of Theorem 6.1)**.** *Composing the bijections*

$$\{\textit{avoiders of } [n]_2\} \xrightarrow{\textit{Cor. 4.2}} \{(s, \varepsilon) : \varepsilon \textit{ valid for } s\} \xrightarrow{\Phi} W_n$$

(the middle set ranges over all Dyck words, but by Lemmas 5.2 and 5.3 only admissible shapes carry valid sign words, so the middle set is exactly the domain of Φ) and applying Lemma 7.2 gives $c_n = |W_n| = 3^n - 3 \cdot 2^{n-1} + 1$. □

The end-to-end map is explicit and runs in linear time in both directions: given an avoider w, read off (s, p) (Lemma 2.1), convert p to ε (Lemma 3.2), and write σ (Definition 7.3); given $\sigma \in W_n$, parse it (Theorem 7.4, Step 2), rebuild s from the parameters (Lemma 5.3), rebuild p from ε (Lemma 3.2), and interleave (Lemma 2.1).

Example 7.6. For $n = 3$, the 16 avoiders and their codes $\sigma = \Phi(\,\cdot\,)$ are:

σ	word	case	σ	word	case
AAA	321321	S	ACB	221133	I, $a=1$
AAB	213213	S	ACC	221313	I, $a=1$
AAC	323211	I, $a=2$	BAC	232311	I, $a=2$
ABA	231231	S	BBC	121233	I, $a=2$
ABB	123123	S	BCA	223311	I, $a=1$
ABC	212133	I, $a=2$	BCB	112233	I, $a=1$
ACA	332211	I, $a=1$	BCC	223131	I, $a=1$
			CCA	232131	II
			CCB	212313	II

The $3 \cdot 2^2 - 1 = 11$ excluded strings are $X_1 = \{\mathsf{BAA}, \mathsf{BAB}, \mathsf{BBA}, \mathsf{BBB}\}$ and $X_2 = \{$CAA, CAB, CBA, CBB, CAC, CBC, CCC$\}$.

Example 7.7. A case (II) example with $n = 6$: $\sigma = \mathsf{CBCCAC}$. Parsing: $\sigma_1 = \mathsf{C}$; the $\{\mathsf{A}, \mathsf{B}\}$-run from position 2 is "B", so $i = 2$ and $\varepsilon_2 = \mathsf{H}$; the C-run occupies positions $3, 4$, so $a = 4$; $\varepsilon_5 = \ell^{-1}(\mathsf{A}) = \mathsf{L}$;

the block gives $\varepsilon_3 = \varepsilon_4 = \overline{\mathsf{L}} = \mathsf{H}$; the tail arc 6 has $\sigma_6 = \mathsf{C}$, so $\delta_6 = 1$ and $\varepsilon_6 = \overline{\varepsilon_5} = \mathsf{H}$. The shape is $s = \mathsf{U}^4\mathsf{D}^2\mathsf{U}\mathsf{D}^2 \cdot \mathrm{Tail}(1;\,1) = \mathsf{UUUUDDUDDUDD}$, the sign word $(\mathsf{H},\mathsf{H},\mathsf{H},\mathsf{L},\mathsf{H})$ gives $p = (2,3,4,5,1,6)$, and interleaving yields

$$w \;=\; 2\,3\,4\,5\,2\,3\,1\,4\,5\,6\,1\,6.$$

One checks directly from the definitions that w is a nonnesting avoider of $\{1132, 3312\}$ and that Φ maps it back to CBCCAC.

Remark 7.8. The encoding explains the shape of the formula bijectively: each position of σ records a three-way choice — "new low" (A), "new high" (B), or "sign forced by the geometry" (C) — for an associated arc, with the position-to-arc correspondence of Definition 7.3 (position j describes arc j, except that in case (I) the first-run letters are shifted one slot left), and with position 1 absorbing the case distinction in cases (S) and (II). The two excluded families are exactly the codes that would describe nothing: a C-free code must describe the staircase, whose first letter carries no information (making the choice B redundant: X_1), and a leading C promises a gap opener, which forces a nonempty constant block *and* a later free arc; the codes $\mathsf{C}v\mathsf{C}^k$ lack one or the other (X_2).

A Machine verification

The conjecture was checked by its authors for $n \le 8$ [5, Table 4]. Independently of the proofs in this paper, every lemma above has been verified mechanically, in most cases far beyond that range, by multiple independently written programs (Python and Rust, plus clean-room re-implementations written by a different model family during adversarial review; see Appendix B). Ground truth for $n \le 8$ was established by three independent enumerators working from the literal containment definition of Section 2.1 and cross-validated against each other, giving

$$(c_n)_{n=1}^{8} = 1,\ 4,\ 16,\ 58,\ 196,\ 634,\ 1996,\ 6178,$$

in exact agreement with $3^n - 3 \cdot 2^{n-1} + 1$, and nonnesting totals $n!\,C_n = 1$, 4, 30, 336, 5040, 95040, 2162160, 57657600. Ground-truth avoider *lists* for $n = 9, 10$ (18,916 and 57,514 words — two levels beyond the conjecture's original range) were generated by a pruned depth-first search that is independent of the shape classification of Section 5. The table below summarizes the principal checks.

Claim checked	Scope	Result
Fast nonnesting/avoidance tests ≡ literal biconditional containment	all 113,400 words of $[5]_2$ and all words for $n \le 4$	0 mismatches
Theorem 4.1, word level: predicted $\iff$ actual avoider, every (shape, permutation) pair	all $n \le 7$; 2,162,160 pairs at $n = 7$	0 mismatches
Theorem 4.1 at $n = 8$, word level, full exhaustive	all 57,657,600 pairs (clean-room Rust harness written by the adversarial referee)	0 mismatches; 6,178 avoiders both ways
Randomized hunts beyond the exhaustive range	$n = 9, 10, 11$: 2,000,000 uniform (shape, permutation) samples each, plus 2,000,000 (shape, prefix-interval permutation) samples each	0 mismatches
Classification (Lemmas 5.2–5.3): (C) satisfiable $\iff$ admissible; per-shape count $= 2^{\lvert F(s)\rvert}$	brute force over all 2^{n-1} sign words per shape, $n \le 9$; structural checks for every Dyck shape $n \le 13$ (742,900 shapes at $n = 13$)	exact equality
Per-shape counts vs. independent Rust per-shape ground truth	all 1,430 shapes at $n = 8$	exact match
$\sum_s N(s)$ over all Dyck shapes, with $N(s) = 0$ for inadmissible s, $= 3^n - 3 \cdot 2^{n-1} + 1$	$n \le 12$ (525,298 at $n = 12$, matching A168583)	pass
Summation identity of Theorem 6.1 as exact integer identity	$n \le 60$	pass
$\lvert W_n\rvert$ and Definition 7.1 ≡ complement description (Lemma 7.2)	full 3^n cube, $n \le 10$	pass
Bijection Φ: well-defined, injective, image $= W_n$, $\Psi \circ \Phi = \mathrm{id}$, $\Phi \circ \Psi = \mathrm{id}$	every valid pair $n \le 8$; end-to-end on all 85,513 ground-truth avoiders for $n \le 10$ (incl. the independent DFS lists at $n = 9, 10$)	pass

In addition, the proofs went through six independent adversarial audits (three for the route of Sections 5–6, three for the bijection of Section 7), each combining line-by-line hand verification of every inference with a fresh clean-room verification suite, and each including a hostile-referee round by a model from a different family with full code-execution access; all six returned the verdict *sound*, with no mathematical gap found. The worked examples (Examples 4.3, 7.6 and 7.7) were re-derived directly from the literal containment definition.

The small-n layer of this table can be reproduced from scratch with the self-contained script `_recompute.py` that accompanies this paper: it implements only the literal definitions and the statements of the lemmas (no code is shared with the provers or the earlier verification suites) and re-checks, among other things: the avoider counts for $n \le 5$ over all words of $[n]_2$; Theorem 4.1 for all pairs with $n \le 6$ in the default run, and with $n \le 7$ under the optional flag `--n7`; the per-shape counts $N(s) = 2^{\lvert F(s)\rvert}$ by brute force over all sign words for $n \le 9$, and the classification totals for $n \le 12$; the language W_n for $n \le 10$; the bijection for $n \le 8$; and all three worked examples (`ALL CHECKS PASS`; on the preparation machine, under CPython 3.11, about fifteen seconds for the default suite and a few seconds for the $n = 7$ sweep). The heavier artifacts quoted in the table above — the clean-room Rust harness used for the exhaustive $n = 8$ sweep (recompiled from source and re-run during the final referee pass: 57,657,600 pairs, 0 mismatches), the shape-level scripts run to $n = 13$, the DFS-generated avoider lists for $n = 9, 10$, and the audit logs — are preserved in the verification bundle that accompanies the source of this paper.

B Methods disclosure

This paper, including all mathematical content, was produced by an artificial-intelligence workflow, and we believe readers and referees should be able to take this fact into account. The conjecture was selected, attacked, and solved by language-model agents (Anthropic's Claude family) running in Demonstrandum, an orchestrated, verification-first multi-agent pipeline: the conventions of Section 2.1 were pinned, as verbatim quotations of the published LaTeX source of [5], in a frozen definitions file before any proof attempt; six proof drafts were generated by parallel agents, of which two survived triage and were merged into the present exposition (the summation endgame of Section 6 and the bijection of Section 7 were found by different agents and are logically independent); and the drafts were then subjected to the six adversarial audits described in Appendix A. A model from a different family (OpenAI's GPT-5.5, accessed through the Codex CLI with code-execution access) acted throughout as a hostile referee: it was prompted to find counterexamples and gaps rather than to confirm, it wrote its own clean-room verification code (including the exhaustive $n = 8$ Rust check of Appendix A), and its objections — all presentational — were incorporated. No human mathematician contributed mathematical content; the human role was limited to operating the pipeline.

We emphasize the epistemic status of the result: correctness rests on the proofs printed in this paper, which are elementary, finite, and intended to be checkable line by line by a human referee in the ordinary way. The machine verification of Appendix A is corroboration, not a substitute for the proofs; conversely, no step of the proofs depends on unpublished computation. Every number quoted in this paper was either recomputed for this manuscript by the accompanying script, recomputed during the final hostile-referee passes on the finished manuscript, or copied from the verification logs of the audits described above. The proofs were written with eventual formalization in a proof assistant in mind (all objects are finite words and tuples; all maps are given explicitly in both directions), but no formalization is claimed here.

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